ALA-1

Code With Mistake Comments

```
print("Exam Result Analyzer")
subjects = int(input("Enter number of subjects: "))
i = 0
total = 0
fail_count = 0
while i < subjects:  # Loop starts
marks = int(input("Enter marks: "))  # ✕ Indentation error
total = total + marks  # ✕ Indentation error
if marks < 35:  # ✕ Indentation error
fail_count = fail_count + 1  # ✕ Indentation error
i = i + 1  # ✕ Indentation error
percentage = total / subjects  # ✕ No zero check (may cause ZeroDivisionError)
print("Total:", total)
print("Percentage:", percentage)
print("Fail count:", fail_count  # ✕ Missing closing parenthesis
if fail_count > 0:
print("Result: Fail")  # ✕ Indentation error
elif percentage >= 75:
print("Distinction")  # ✕ Indentation error
elif percentage >= 60:
print("First Class")  # ✕ Indentation error
else:
print("Pass")  # ✕ Indentation error
if percentage > 90  # ✕ Missing colon
print("Excellent performance")
else:
print("Keep improving")
```

Mistake Explanation

1. **Indentation Error** – Code inside while and if blocks was not indented properly. Python requires indentation to define blocks.

2. **Missing Parenthesis** – One print() statement was missing a closing) which caused a SyntaxError.

3. **Missing Colon** – if percentage > 90 was missing : at the end.

4. **Zero Division Risk** – percentage = total / subjects was written without checking if subjects == 0, which could cause a ZeroDivisionError.

Correct Code

```
print("Exam Result Analyzer")
try:
   subjects = int(input("Enter number of subjects: "))

   if subjects <= 0:
      print("Subjects must be greater than zero.")
   else:
      i = 0
      total = 0
      fail_count = 0

      while i < subjects:
         marks = int(input("Enter marks: "))
         total = total + marks

         if marks < 35:
            fail_count = fail_count + 1

         i = i + 1

      percentage = total / subjects

      print("Total:", total)
      print("Percentage:", percentage)
      print("Fail count:", fail_count)
```

```
    if fail_count > 0:
        print("Result: Fail")
    elif percentage >= 75:
        print("Distinction")
    elif percentage >= 60:
        print("First Class")
    else:
        print("Pass")

    if percentage > 90:
        print("Excellent performance")
    else:
        print("Keep improving")

except ValueError:
    print("Please enter valid numeric input.")
```

output:

```
Run   correct ×

C:\Users\shahh\AppData\Local\Programs\Python\Python314\python.exe C:\Users\shahh\OneDrive\Documents\python\correct.py
Exam Result Analyzer
Enter number of subjects: 6
Enter marks: 60
Enter marks: 70
Enter marks: 80
Enter marks: 90
Enter marks: 70
Enter marks: 65
Total: 435
Percentage: 72.5
Fail count: 0
First Class
Keep improving

Process finished with exit code 0
```